

Database Management System

EXPERIMENT 15 CONTROLLING USER ACCESS

NAME: LUQMAN T

ROLL: 241801141

1. What privilege should a user be given to log on to the Oracle Server? Is this a system or an object privilege?

Answer: CREATE SESSION - This is a system privilege.

2. What privilege should a user be given to create tables?

Answer: CREATE TABLE - This is a system privilege.

3. If you create a table, who can pass along privileges to other users on your table?

Answer: The table owner or users who have been granted the privilege WITH GRANT OPTION.

4. You are the DBA. You are creating many users who require the same system privileges. What should you use to make your job easier?

Answer: Create a ROLE and grant the system privileges to the role, then grant the role to multiple users.

5. What command do you use to change your password?

Answer: ALTER USER username IDENTIFIED BY new_password;

6. Grant another user access to your DEPARTMENTS table. Have the user grant you query access to his or her DEPARTMENTS table.

As DBA or table owner:

```
-- Grant SELECT privilege on DEPARTMENTS to another user
GRANT SELECT ON departments TO user2;
```

Expected Output:

Grant succeeded.

As the other user (user2), grant SELECT back:

```
GRANT SELECT ON departments TO original_user;
```

Expected Output:

Grant succeeded.

7. Query all the rows in your DEPARTMENTS table.

```
SELECT * FROM departments;
```

Expected Output:

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID
10	Administration	200
20	Marketing	201

1800		
	30 Purchasing	114
1700		
	40 Human Resources	203
2400		
	50 Shipping	121
1500		
	60 IT	103
1400		
	70 Public Relations	204
2700		
	80 Sales	145
2500		
	90 Executive	100
1700		
	100 Finance	108
1700		
	110 Accounting	205
1700		
	120 Treasury	-
1700		
	130 Corporate Tax	-
1700		
	140 Control And Credit	-
1700		
	150 Shareholder Services	-
1700		
	160 Benefits	-
1700		
	170 Manufacturing	-
1700		
	180 Construction	-
1700		
	190 Contracting	-

1700	200 Operations	-
1700	210 IT Support	-
1700	220 NOC	-
1700	230 IT Helpdesk	-
1700	240 Government Sales	-
1700	250 Retail Sales	-
1700	260 Recruiting	-
1700	270 Payroll	-

8. Add new rows to DEPARTMENTS table

Team 1:

```
INSERT INTO departments (department_id, department_name,
location_id)
VALUES (500, 'Education', 1700);
```

Expected Output:

1 row created.

Team 2:

```
INSERT INTO departments (department_id, department_name,  
location_id)  
VALUES (510, 'Human Resources', 1700);
```

Expected Output:

1 row created.

Query other team's table (if privileges granted):

```
-- If granted SELECT privilege on the other team's table  
SELECT * FROM team2.departments WHERE department_id = 510;
```

G. Query USER_TABLES data dictionary

```
SELECT table_name, tablespace_name, num_rows  
FROM user_tables  
WHERE table_name = 'DEPARTMENTS';
```

Expected Output:

TABLE_NAME	TABLESPACE_NAME
DEPARTMENTS	USERS
27	

10. Revoke SELECT privilege from the other team

```
REVOKE SELECT ON departments FROM user2;
```

Expected Output:

Revoke succeeded.

11. Remove inserted row and save changes

```
DELETE FROM departments WHERE department_id = 500;  
COMMIT;
```

Expected Output:

1 row deleted.

Commit complete.

Complete User Management Examples:**1. Create a new user**

```
-- As DBA  
CREATE USER new_user IDENTIFIED BY password123;
```

Expected Output:

User created.

2. Grant system privileges to user

```
GRANT CREATE SESSION, CREATE TABLE, CREATE VIEW TO new_user;
```

Expected Output:

Grant succeeded.

3. Grant object privileges with GRANT OPTION

```
GRANT SELECT, INSERT, UPDATE ON employees TO new_user WITH  
GRANT OPTION;
```

Expected Output:

Grant succeeded.

4. Create and use roles

```
-- Create role  
CREATE ROLE manager_role;  
  
-- Grant privileges to role  
GRANT CREATE TABLE, CREATE VIEW, CREATE PROCEDURE TO  
manager_role;  
  
-- Grant role to user  
GRANT manager_role TO new_user;
```

Expected Output:

Role created.

Grant succeeded.

Grant succeeded.

5. View user privileges

```
-- View system privileges
SELECT * FROM user_sys_privs;

-- View role privileges
SELECT * FROM user_role_privs;

-- View object privileges
SELECT * FROM user_tab_privs;
```

Expected Output (sample):

USERNAME	PRIVILEGE
ADM	

NEW_USER	CREATE SESSION
NO	
NEW_USER	CREATE TABLE
NO	
NEW_USER	CREATE VIEW
NO	

6. Change user password

```
ALTER USER new_user IDENTIFIED BY newpassword456;
```

Expected Output:

User altered.

7. Grant privileges to PUBLIC


```
GRANT SELECT ON countries TO PUBLIC;
```

Expected Output:

Grant succeeded.

8. Revoke privileges

```
REVOKE INSERT ON employees FROM new_user;
```

Expected Output:

Revoke succeeded.

G. Drop user

```
-- If user has objects  
DROP USER new_user CASCADE;
```

```
-- If user has no objects  
DROP USER new_user;
```

Expected Output:

User dropped.

Security Best Practices Examples:

1. Create a read-only role

```
CREATE ROLE read_only_role;  
GRANT CREATE SESSION TO read_only_role;  
GRANT SELECT ON employees TO read_only_role;
```

```
GRANT SELECT ON departments TO read_only_role;  
GRANT read_only_role TO report_user;
```

2. Create an application role

```
CREATE ROLE hr_app_role;  
GRANT SELECT, INSERT, UPDATE ON employees TO hr_app_role;  
GRANT SELECT, INSERT, UPDATE ON departments TO hr_app_role;  
GRANT SELECT ON jobs TO hr_app_role;  
GRANT SELECT ON locations TO hr_app_role;
```

3. View all privileges in the system

```
-- View all users  
SELECT username, account_status, created FROM dba_users;  
  
-- View all roles  
SELECT role, password_required FROM dba_roles;  
  
-- View role privileges  
SELECT grantee, granted_role FROM dba_role_privs;
```

Key Security Concepts Demonstrated:

1. **System Privileges** - Database operations (CREATE TABLE, CREATE SESSION, etc.)
2. **Object Privileges** - Operations on specific objects (SELECT, INSERT, UPDATE, DELETE)
3. **Roles** - Named groups of privileges for easier management
4. **GRANT** - Gives privileges to users or roles
5. **REVOKE** - Removes privileges from users or roles
6. **WITH GRANT OPTION** - Allows grantee to grant privileges to others
7. **PUBLIC** - Special group that includes all users

8. Data Dictionary Views - USER_TAB_PRIVS, USER_SYS_PRIVS, USER_ROLE_PRIVS